

Briefing Note

Title: Advice to Minister of Health, summary of project on transmission dynamics of influenza in 2016-2017 season.

Issue: Minister of Health requires a report to summarize the project on transmission dynamics of influenza within the Albertan population from the Math572 final project. Summary of a study about influenza dynamics of Albertans from 2016-2017 is required.

Purpose: To provide a summary of a final project modelling the influenza dynamics in Alberta.

Background:

- A request was sent Dec 2022 and was granted permission to take MATH 572 course.
- New summary of the final project on transmission dynamics of influenza from the course is required.
- Summary of final project: "Modelling Seasonal Influenza Data in Alberta" is required.

Analysis

- Mathematical model developed to observe epidemiological patterns of influenza from 2016-2017 to explore the potential impact of preventative measures such as recommending increase of vaccine coverage.
- Maximizing vaccine effectiveness and coverage of the population decreases cumulative cases of children up to 45%.
- Due to underreporting of influenza, model only captures fraction of the actual population.

Stakeholder Reaction:

- External stakeholders may request policy on increasing vaccine coverage to reduce spread of influenza within the Albertan community.
- The Ministry of Health knows the impact of vaccines for influenza and may allocate more funding towards vaccine research and inform the department of health to allocate more efforts in increasing vaccination coverage.

Communications/Key Messages:

- Vaccine effectiveness and coverage has major impacts on reducing cases of influenza.
- Additional studies in Alberta that model various seasons of influenza within the population is required.
- The Ministry of Health continues to prioritize its effort to reduce number of influenza cases in the Alberta's health industry.

Financial Implications: NONE

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